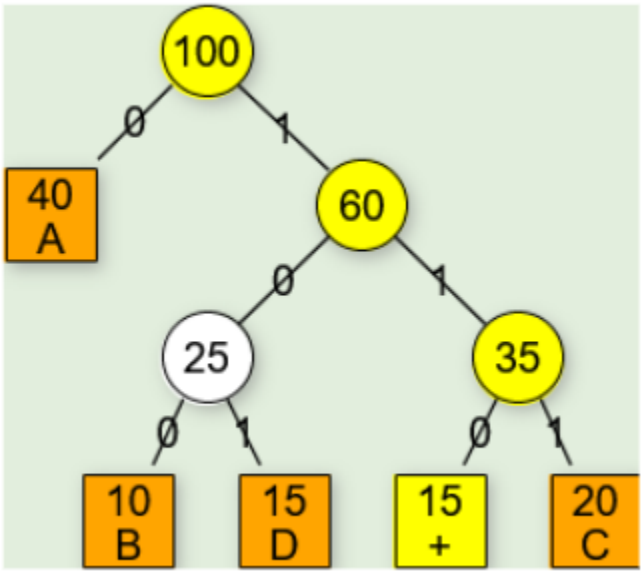
(a) You are given the following table containing symbols and their frequencies:

| **Symbol** | A | B | C | D | + |
|---|---|---|---|---|---|
| **Frequency** | 40 | 10 | 20 | 15 | 15 |

i) Build the Huffman code tree and find the codeword for each character.
ii) Decode 100010111001010 using the Huffman code that you generated.

**Huffman Tree**

i)




ii) Decode: **BAD+ADA**

A team of two infamous thieves, Denver and Nairobi, planned to rob the famous Louvre Museum. Before the scene, they both agreed on the fact that none of them will break any item as all the items in the Louvre are too precious, and taking a fraction of any item won't sell in the black market. If it fits in the bag as a whole, they will take it, otherwise, leave it as it is.

Both of them arrived at the Louvre with an empty knapsack weighing a total of 8 kg. Despite the fact that both thieves are experts in their fields, they take slightly different approaches. Denver believes he will use a Dynamic Programming Approach to rob the items in the most efficient manner possible. Nairobi, on the other hand, believes that if she chooses a Greedy Approach, she will make the most money.

The objects in the Louvre Museum are listed below.

| **Objects** | **Jewelry** | **Sculpture** | **Painting** | **Book** | **Mummy** |
|---|---|---|---|---|---|
| **Profit ($)** | 5 | 9 | 5 | 4 | 6 |
| **Weight (Kg)** | 3 | 5 | 4 | 1 | 12 |

(a) What is the maximum profit Denver can make using his strategy? What items did he pick up? Show how Denver used the DP table to select these objects. You may use arrows and circles to point to the chosen cells.
(b) Does Nairobi's belief remain valid after the robbery? Prove it.

a) Best value Denver can achieve is 14 $ (Jewelry and Sculpture)

Knapsack Table

| | 0 | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|---|---|
| 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 | 0 |
| 1 | 0 | 0 | 0 | 5 | 5 | 5 | 5 | 5 | 5 |
| 2 | 0 | 0 | 0 | 5 | 5 | 9 | 9 | 9 | 14 |
| 3 | 0 | 0 | 0 | 5 | 5 | 9 | 9 | 10 | 14 |
| 4 | 0 | 4 | 4 | 5 | 9 | 9 | 13 | 13 | 14 |
| 5 | 0 | 4 | 4 | 5 | 9 | 9 | 13 | 13 | 14 |

b) For Nairobi:

| **Objects** | **Jewelry** | **Sculpture** | **Painting** | **Book** | **Mummy** |
|---|---|---|---|---|---|
| $Profit/Weight$ | 1.67 | 1.8 | 1.25 | 4 | 0.5 |

| **Weight Remaining** | **Object Taken** | **Object Weight** | **Object Profit** |
|---|---|---|---|
| **8** | **Book** | **1** | **4** |
| **7** | **Sculpture** | **5** | **9** |
| **2** | **No Item can be taken according to thieves' agreement** | | |
| **Total Profit:** | **4 + 9 = 13 $** | | |

So, No, Nairobi's belief is not true in this approach.

Even if Nairobi took Jewelry and Sculpture (3 kg + 5 kg = 8 Kg), She couldn't make more profit than Denver. In this case, her profit would be 14 $ too.